BEST BIAS DISCOVERY · ZINDI BIAS BOUNTY: MAPPING EQUITY

A third of fire stations are missing from Overture, most of all in wildfire country and Oklahoma's tribal areas

Zindi user **mrSharafi** · 1 November 2026 · data and code: see "Method, data and reproducibility"

On the Klamath River in Humboldt County, California, the Yurok Volunteer Fire Department runs Station 1 in Weitchpec. Its tract's wildfire hazard index is 2,233; the top-quarter cut-off in this study is 153.3. Overture's map has no fire station within a kilometre of it. In southeastern Oklahoma, inside the Choctaw Nation, the Battiest and Pickens volunteer departments are missing too. Across four regions, 1,400 of 4,625 public-record stations are missing, and a typical missing station is 6.3 km from the nearest fire place Overture does show.

30.3%
of 4,625 USGS fire stations missing from Overture (schools: 1.8%)

57.5%
missing in tribal-area tracts with top-quarter wildfire hazard (245 of the 285 are in eastern Oklahoma)

6.3 km
median distance from a missing station to the nearest fire place Overture shows

84.2%
of Overture's fire places come from Meta alone (all places: 60.4%)

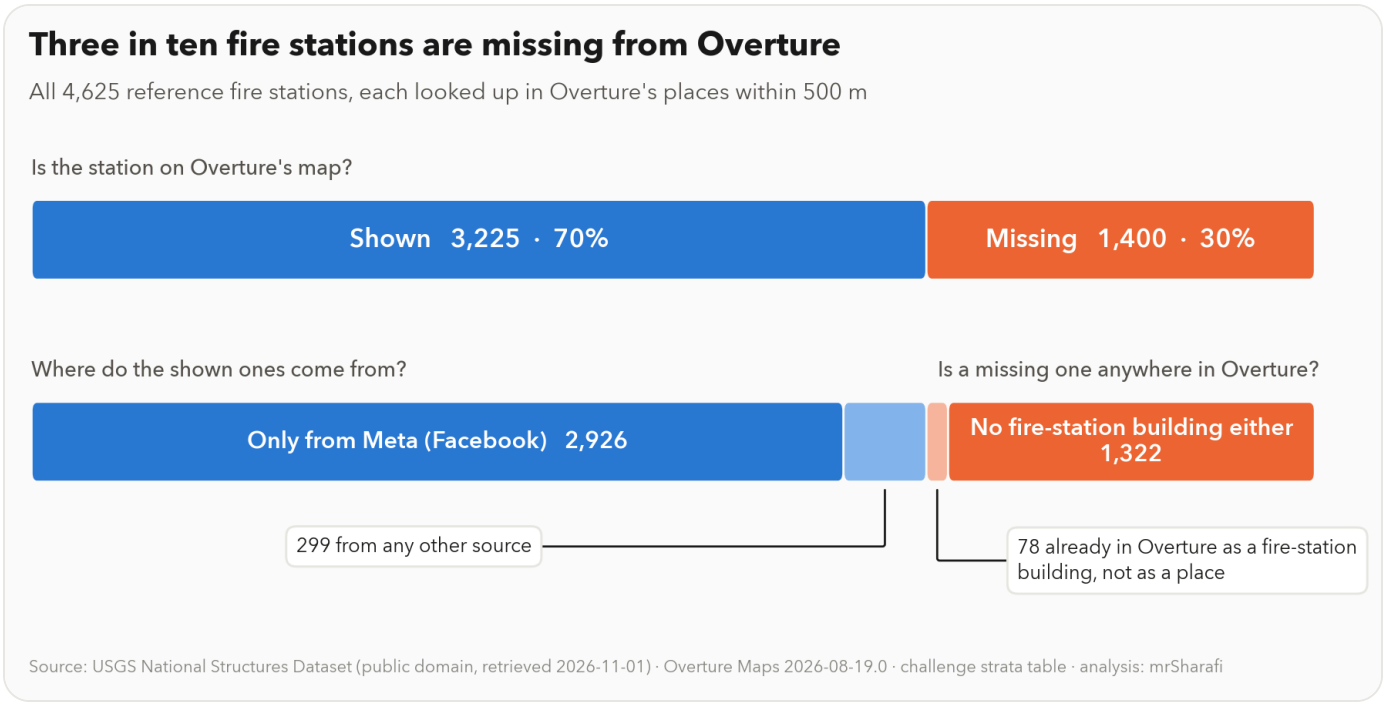


Fig. 1. Each of the 4,625 stations in the public USGS National Structures Dataset, looked up in Overture's places (release 2026-08-19.0). Top row: is the station on the map. Bottom row: where the shown ones come from, and whether a missing one has at least a fire-station building in Overture.

Who is affected

The missing stations are not slightly misplaced; they are absent from their whole neighbourhood. The median missing station is 6.3 km from the nearest fire place Overture shows, and 58.8% are more than 5 km from one. In rural tribal-area tracts the median is 8.9 km; in urban non-tribal tracts it is 2.6 km. 536 missing stations sit in top-quarter-hazard tracts; those 289 tracts have 1,065,798 residents (tract totals, not people left without a station).

The named cases are the Yurok and Hoopa Valley country on the Klamath and Trinity rivers and the Anderson Valley in Mendocino County, the highest-hazard tracts in the study's named list; in Oklahoma, the Choctaw Nation (Battiest, Pickens, Honobia, Finley), the Cherokee Nation (Stilwell) and Osage County (Big Bend, Drum Creek). Fire stations are where engines stage for wildland response and where residents go when roads close. Any station finder, routing or coverage-gap analysis built on Overture shows no station at these addresses, and every nearest-station query drawn from it skips them.

Finding 1 — Where: wildfire hazard everywhere, tribal areas in Oklahoma

Wildfire hazard and tribal-area status each raise the share of missing stations, and the two stack. Among rural stations, 31.3% are missing in non-tribal lower-hazard tracts, 44.7% in non-tribal top-hazard tracts, 46.8% in tribal-area lower-hazard tracts and 59.1% in tribal-area top-hazard tracts (n = 269).

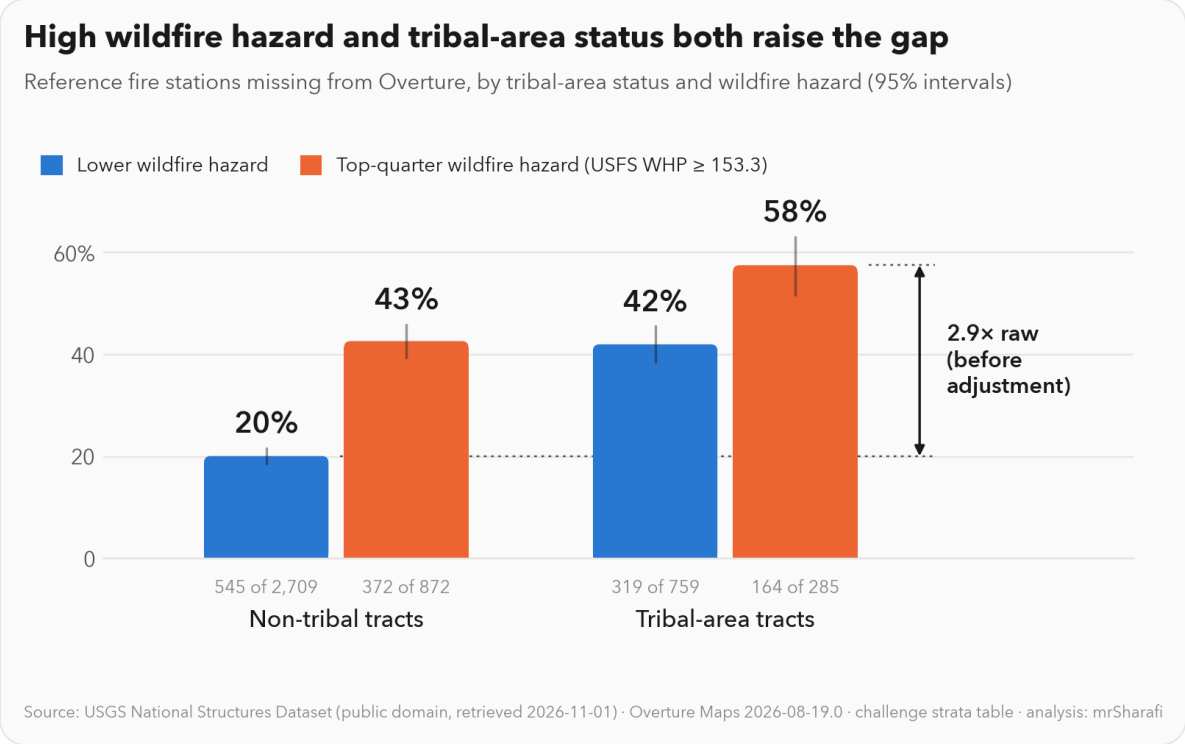


Fig. 2. All stations, before any adjustment: 164/285 = 57.5% missing in tribal-area top-hazard tracts against 545/2709 = 20.1% in non-tribal lower-hazard tracts; the 2.9× arrow is that raw ratio. Whiskers are 95% Wilson intervals. Top-quarter hazard = tract USFS Wildfire Hazard Potential of 153.3 or more, the 75th percentile across stations.

Is this just Oklahoma, or just rurality? Top-quarter hazard raises the missing share in all four regions: eastern Oklahoma 60.5% vs 38.7%, northern California 54.1% vs 16.9%, Maricopa 29.9% vs 19.3%, south-central Texas 31.3% vs 20.6%. Holding region, rural/urban and tribal-area status fixed, a station's odds of being missing are

1.62 times higher in top-hazard tracts (95% CI 1.36–1.93). The tribal-area effect is smaller and regional: with region, rural/urban and hazard fixed the odds ratio is 1.45 (1.04–2.02). It is clear in eastern Oklahoma (46.5% vs 26.3%; odds ratio 2.28, 1.54–3.36) and Arizona (42.4% vs 22.4%, only 33 tribal-area stations), but absent in northern California (41.7% vs 46.1%, 24 stations). Without the rural/urban adjustment the odds ratios are 2.29 and 1.91.

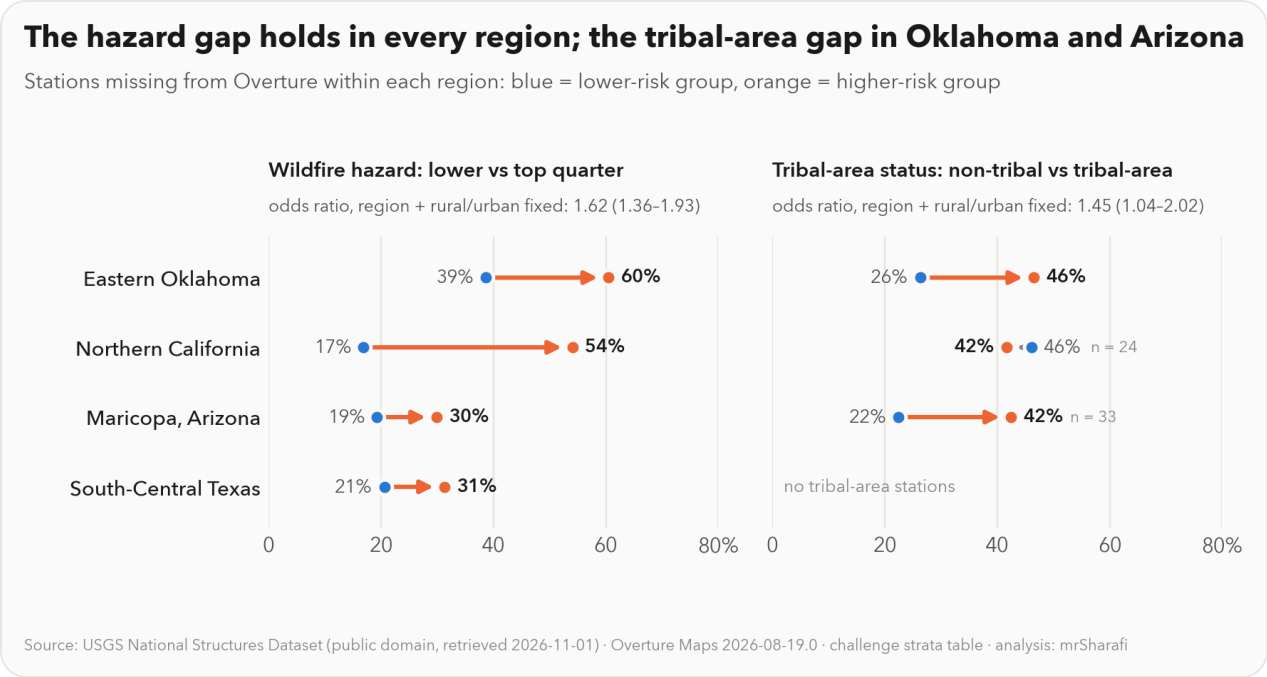


Fig. 3. Within-region shares; the odds ratios hold region, rural/urban and the other factor fixed (see Method).

Finding 2 – Why: fire stations, not facilities in general, and a feed with no backup

Overture does not lose public facilities in general. Under the same 500 m test, 1.8% of 12,585 USGS schools and 1.0% of 768 hospitals are missing, against 30.3% of fire stations. What sets fire stations apart is where Overture gets them: 84.2% of Overture's fire places have Meta as their only upstream source, against 78.2% of civic places and 60.4% of all places; among the stations Overture does show it is 90.7%. Meta dependence alone does not explain the gap, since civic places are almost as Meta-heavy and are rarely missing. What it does explain is fragility: where Meta has no fire-station record, Overture has almost no second route to one. We cannot see Meta's own data, so this is a well-supported mechanism, not a proven cause.

Yet for **78** of the 1,400 missing stations, Overture already holds the station: its own buildings theme has a footprint of class `fire_station` (from OpenStreetMap) within 500 m. Linking those footprints to the places theme would restore them with data Overture already ships (the pale orange slice in Fig. 1).

Where the missing stations are

The missing stations (orange) spread across the countryside, while the stations Overture shows (blue) cluster in the cities. Most of eastern Oklahoma is tribal-area land (shaded):

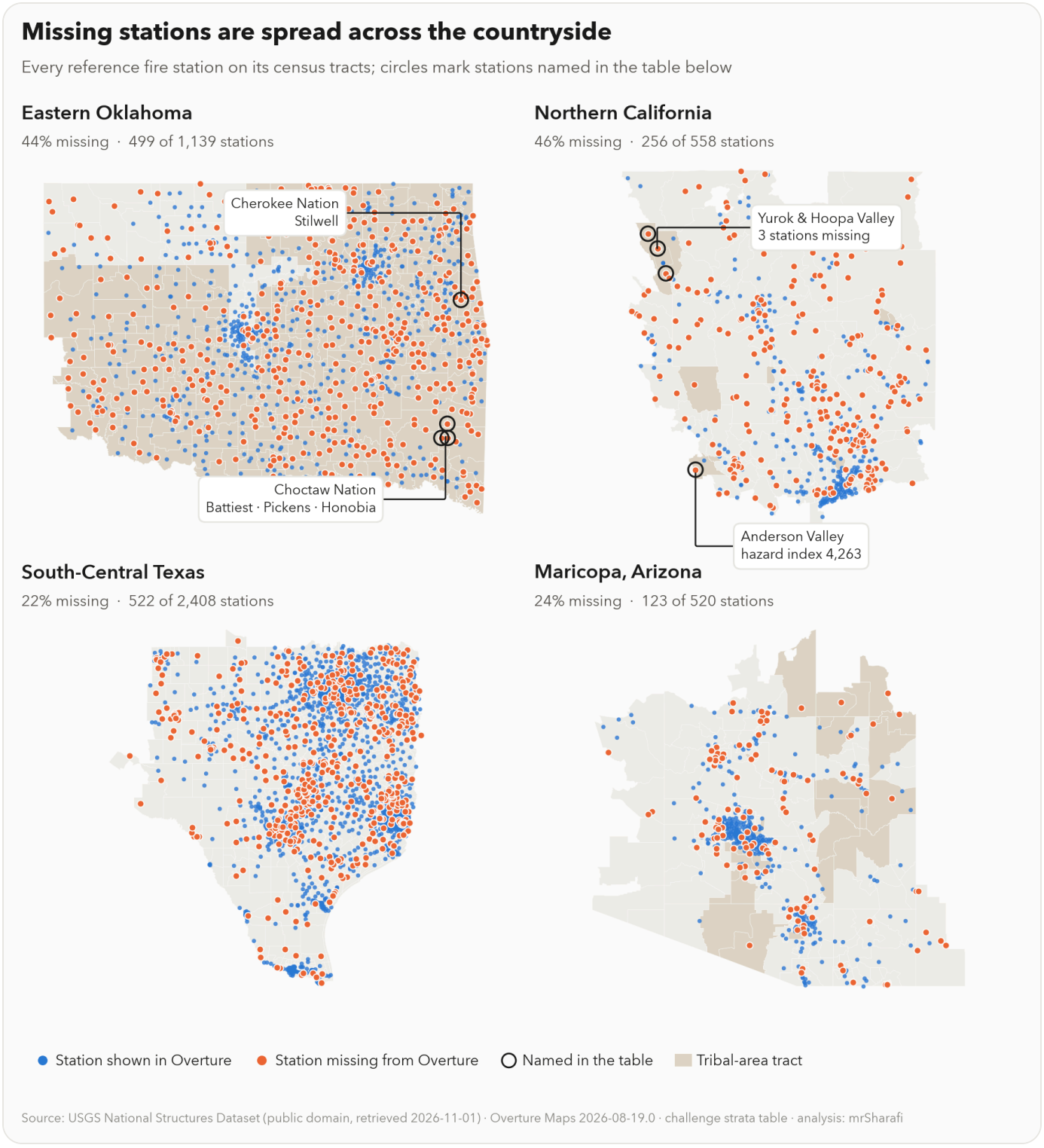


Fig. 4. Tribal-area tracts are shaded (`tribal_any` in the challenge's strata table). Circles mark stations named in the named-stations table.

Up close the pattern is stark. Along the Klamath and Trinity rivers, Overture shows Orleans and the two Hoopa stations, but not Yurok Station 1, Yurok Station 2, Willow Creek, Salmon River or Salyer. In southeastern Oklahoma, inside the Choctaw Nation, most of the stations in the frame are missing:

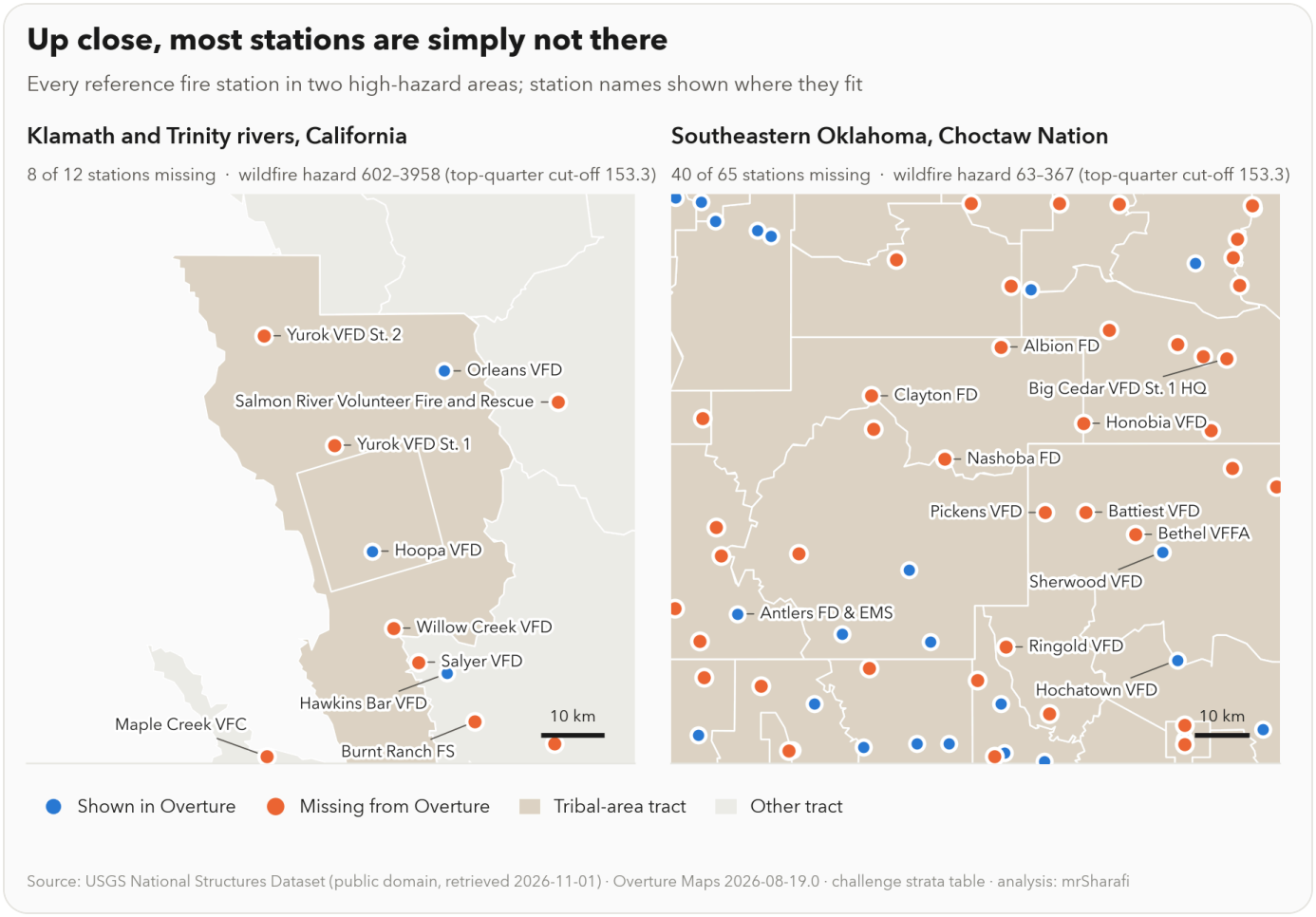


Fig. 5. The frames are about 95 km (California) and 110 km (Oklahoma) across; counts in the panel headings cover every reference station inside the frame. Frames were chosen in high-hazard areas and are more affected than their regional averages.

Finding 3 — It is the place, not the kind of department

Volunteer stations are missed more overall (35.2% vs 27.9%), but that follows from where they are. Within each tract type the two are missed at similar rates: rural tribal-area 52.1% vs 49.6%. In rural non-tribal tracts volunteer stations are missed *less* (33.5% vs 41.4%). The data do not support a "volunteer departments have no Facebook page" explanation (we cannot observe Facebook pages directly).

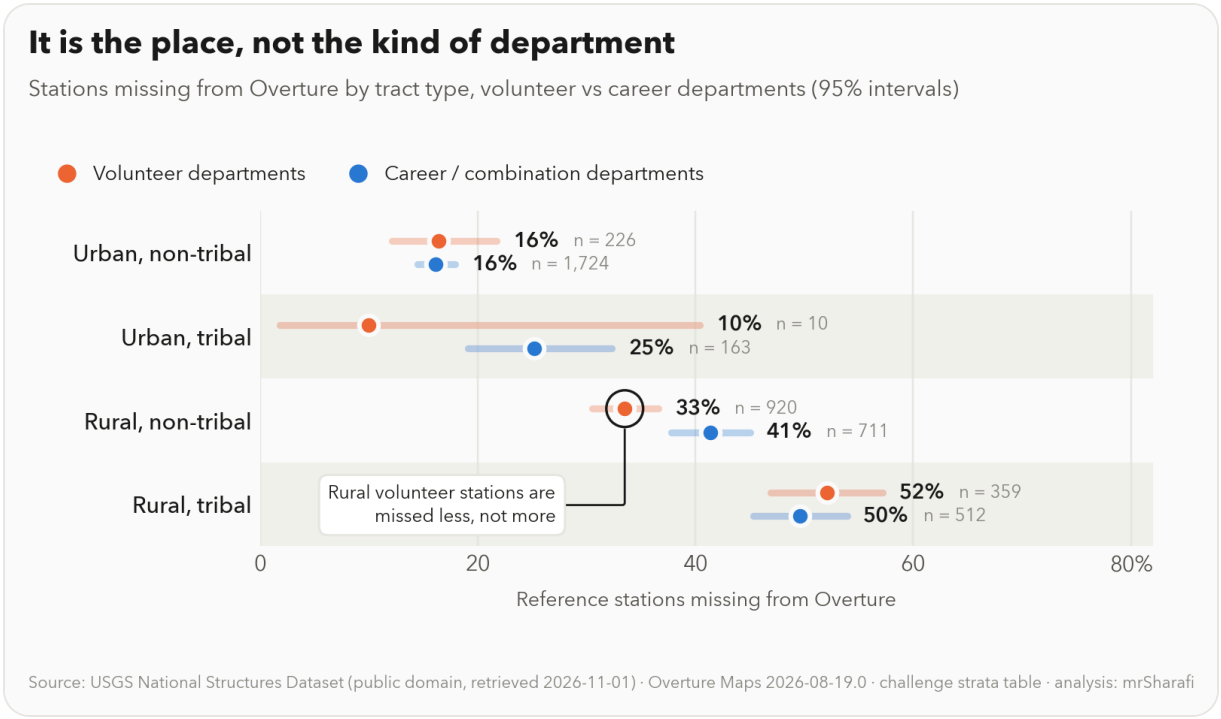


Fig. 6. Lines are 95% Wilson intervals; n is the number of reference stations in each group. Cell counts are in the appendix.

How strong is this?

check	result
Matching radius	missing shares move by under 2 points at 250 m and 1 km (volunteer 37.0% / 35.2% / 33.7%; other 29.2% / 27.9% / 26.3%)
Mislocation	the median missing station is 6.3 km from any Overture fire place; 58.8% are more than 5 km away
Generic rural sparsity	same test: schools 1.8% missing (n = 12,585), hospitals 1.0% (n = 768)
Rurality	hazard odds ratio 1.62 (95% CI 1.36–1.93), tribal-area 1.45 (95% CI 1.04–2.02), with region and rural/urban held fixed
Region	hazard gap in all four regions; tribal-area gap in Oklahoma and Arizona, not northern California
Provenance baseline	Meta-only: fire places 84.2%, civic 78.2%, all places 60.4%
Public re-run	4,625 stations from the live USGS service; all 4,625 match the earlier challenge-package copy (3 stations it listed are no longer in the service)

Limitations.

- Matching is by distance and fire category, not by name, so errors run both ways: a nearby unrelated fire place can make a station look present, and a station filed under a non-fire name in a shared civic building can make it look missing.

- The reference can list stations that have closed or moved; we did not verify stations individually.
- Stations in one tract share its hazard value, so intervals that treat stations as independent are somewhat too narrow.

What should change

1. **Overture: add a public second source for emergency facilities.** The USGS National Structures Dataset is public domain and lists every station in this study. Today 84.2% of Overture's fire places rest on Meta alone.
2. **Overture: link what it already ships (a quick win).** 78 of the 1,400 missing stations (5.6%) have an OpenStreetMap `fire_station` building within 500 m in Overture's own buildings theme. Joining those footprints to the places theme restores them with no new data.
3. **Map users: do not trust absence.** In rural and tribal-area tracts, "no station here" in Overture is often wrong. Station finders and coverage analyses should cross-check a public facility list first.
4. **The scorecard: add three views.** A per-facility check against a public reference, with a control category such as schools; a split by data provenance (`sources`); and intersections such as tribal-area × wildfire hazard.

The named stations

12 of the 187 missing volunteer stations in rural tribal-area tracts, highest hazard first (top-quarter cut-off WHP 153.3). The four California rows lead on hazard; their tracts are flagged tribal by overlap and illustrate the hazard effect (in northern California the tribal-area flag does not raise the missing share). None has an Overture fire place within 1 km except Bethel (660 m).

station (reference name)	city	region	tract GEOID	wildfire hazard (USFS WHP)	tract residents
Anderson Valley Volunteer Fire Department Station 744	Yorkville	northern-ca	06045011800	4,263	2,053
Yurok Volunteer Fire Department Station 2	Hoopa	northern-ca	06023010102	2,233	2,914
Yurok Volunteer Fire Department Station 1	Weitchpec	northern-ca	06023010102	2,233	2,914
Willow Creek Volunteer Fire Department	Willow Creek	northern-ca	06023010102	2,233	2,914
Big Bend Volunteer Fire Department	Ralston	eastern-ok	40113940003	384	4,406
Drum Creek Volunteer Fire Department	Ponca City	eastern-ok	40113940003	384	4,406
Honobia Volunteer Fire Department	Honobia	eastern-ok	40127097600	367	2,627

station (reference name)	city	region	tract GEOID	wildfire hazard (USFS WHP)	tract residents
Highway 100 West Volunteer Fire Department	Stilwell	eastern-ok	40001376800	361	3,848
Battiest Volunteer Fire Department	Battiest	eastern-ok	40089098200	356	2,349
Bethel Volunteer Fire Fighters Association	Bethel	eastern-ok	40089098200	356	2,349
Pickens Volunteer Fire Department	Pickens	eastern-ok	40089098200	356	2,349
Finley Community Volunteer Fire Department	Finley	eastern-ok	40127097800	332	2,571

How this differs from other entries

The automated scorecard reports five metrics per tract over fixed strata. This entry works per facility against an independent public list, with control categories (schools, hospitals) showing the omission is specific to fire stations; it splits by data provenance (Overture `sources`, which the scorecard never reads) and crosses tribal-area status with wildfire hazard. None of the entries below reports a per-facility reference check or a control category.

entry	what it shows	what ours adds
Hunter_Toledo (8 Oct), "coverage deserts are authorship deserts"	rural tracts with no Overture fire POI; POIs carry no OSM records	a station-level check against a public reference: real stations, named, that Overture misses, and how far away the nearest shown one is
mugeha-laura (31 Oct), hidden fire departments	fire departments that are in Overture but filed outside the fire categories	the opposite failure: stations absent from Overture altogether. We search fire names in every category, so their "hidden" departments count as shown here
hero76 (31 Oct), ambulances filed as fire stations	wrong records inside Overture's fire category	missing records, checked against the reference
vigohan (14 Oct), phantom roads	unnamed, unedited TIGER-import roads near the Texas border	a different layer (facilities) and a different failure (omission, not phantom features)

Method, data and reproducibility

Method. For each reference station we take the nearest Overture place filed as `fire_department` or `fire_protection_service`, or carrying a fire-station name in any other category. A station is *missing* when no such place lies within 500 m. Tract fields come from the challenge's strata table: rural = `pct_urban` < 0.5; tribal-area = the tract overlaps a reservation or tribal statistical area (`tribal_any`), which in eastern Oklahoma covers most tracts; hazard = `usfs_WHP_mean`, cut at 153.3, the 75th percentile across stations. *Meta-only*: the matched place lists Meta and no other upstream dataset among its sources. Schools and hospitals use the same 500 m rule with their own categories and names. Rates carry 95% Wilson intervals; odds ratios are Mantel–Haenszel, stratified as stated, with Robins–Breslow–Greenland intervals.

Data. Fire stations, schools and hospitals: USGS National Structures Dataset (public domain), National Map service https://carto.nationalmap.gov/arcgis/rest/services/structures/MapServer, layers 16 (Fire Stations/EMS Stations, FCode 74026), 23 (Schools) and 14 (Hospitals/Medical Centers), retrieved 2026-11-01 by `research/fetch_usgs_stations.py`; the snapshots are committed in `data_public/external/`. The live service changes over time, so use the committed snapshot to reproduce the exact numbers. Catalogue: https://catalog.data.gov/dataset/usgs-national-structures-dataset-usgs-national-map-downloadable-data-collection. Overture places and buildings (release 2026-08-19.0), tract polygons and the strata table come from the challenge package (CC BY-SA 4.0). The rules allow additional public datasets for this prize; none of our leaderboard submissions reads the USGS data (`model/paths.py`, `check_compliance.sh`). Associated leaderboard submissions: `RzkTLsFj` and `2gyMPZPa`.

Reproduce (about 3 minutes on a laptop):

```
.venv/bin/python research/fetch_usgs_stations.py              # optional re-fetch
.venv/bin/python research/fetch_usgs_stations.py --layer 23 --out usgs_schools.csv
.venv/bin/python research/fetch_usgs_stations.py --layer 14 --out usgs_hospitals.csv
PYTHONPATH=. .venv/bin/python research/discovery_stations.py  # stations
python3 research/discovery_robust.py                          # robust.json
PYTHONPATH=. .venv/bin/python research/discovery_checks.py    # checks.json
.venv/bin/python research/discovery_meta_baseline.py          # meta baseline
python3 research/discovery_figures_v3.py                      # Figs. 1–6
python3 research/check_entry_numbers.py                       # NUMBERS OK
```

Appendix – full tables

Tribal-area status × wildfire hazard (the cells of Fig. 2; columns: lower hazard, top-quarter hazard).

tract type	not top-quartile wildfire hazard: missing	top-quartile wildfire hazard: missing
non-tribal	545/2709 = 20.1% (18.7-21.7)	372/872 = 42.7% (39.4-46.0)
tribal	319/759 = 42.0% (38.6-45.6)	164/285 = 57.5% (51.7-63.1)

Volunteer vs career by tract type (the cells of Fig. 6).

tract type	volunteer: missing	other: missing
urban, non-tribal	37/226 = 16.4% (12.1-21.8)	278/1724 = 16.1% (14.5-17.9)
urban, tribal	1/10 = 10.0% (1.8-40.4)	41/163 = 25.2% (19.1-32.3)
rural, non-tribal	308/920 = 33.5% (30.5-36.6)	294/711 = 41.4% (37.8-45.0)
rural, tribal	187/359 = 52.1% (46.9-57.2)	254/512 = 49.6% (45.3-53.9)

All stations.

reference stations	n	missing from Overture (no fire place within 500 m)	95% CI
volunteer departments	1515	533 (35.2%)	32.8-37.6%

reference stations	n	missing from Overture (no fire place within 500 m)	95% CI
all other (career/combination)	3110	867 (27.9%)	26.3-29.5%
all	4625	1400 (30.3%)	29.0-31.6%

By region (volunteer stations are missed more in every region; the tract-type table above shows that this follows from where they sit):

region	volunteer: missing	other: missing
eastern-ok	192/384 = 50.0% (45.0-55.0)	307/755 = 40.7% (37.2-44.2)
maricopa-az	12/28 = 42.9% (26.5-60.9)	111/492 = 22.6% (19.1-26.5)
northern-ca	41/78 = 52.6% (41.6-63.3)	215/480 = 44.8% (40.4-49.3)
south-central-tx	288/1025 = 28.1% (25.4-30.9)	234/1383 = 16.9% (15.0-19.0)

BEST DOCUMENTATION · ZINDI BIAS BOUNTY: MAPPING EQUITY

Predicting map coverage gaps without the answer key

Zindi user **mrSharafi** · final submissions `RzkTLsFj` (`c3_clean_regions_grp05` , public RMSE 0.068339) and `2gyMPZPa` (`k84_k55_line145_rown512k3` , public RMSE 0.056774) · 1 November 2026

On 25 September the organisers took away the data their score is computed from. This is how we kept predicting it anyway: keep their formula, estimate what was removed from what is still allowed, and learn the rest from our own leaderboard scores, one exact number at a time.

9,794
census tracts scored, in 5 regions

88
scored uploads, every one logged

0.0568
best public RMSE (k84), down from 0.108

5 s
to rebuild the clean final file (c3), byte for byte

Five steps from permitted data to a scored file

Each step reads only the output of the step before it; the box names the file that does the work

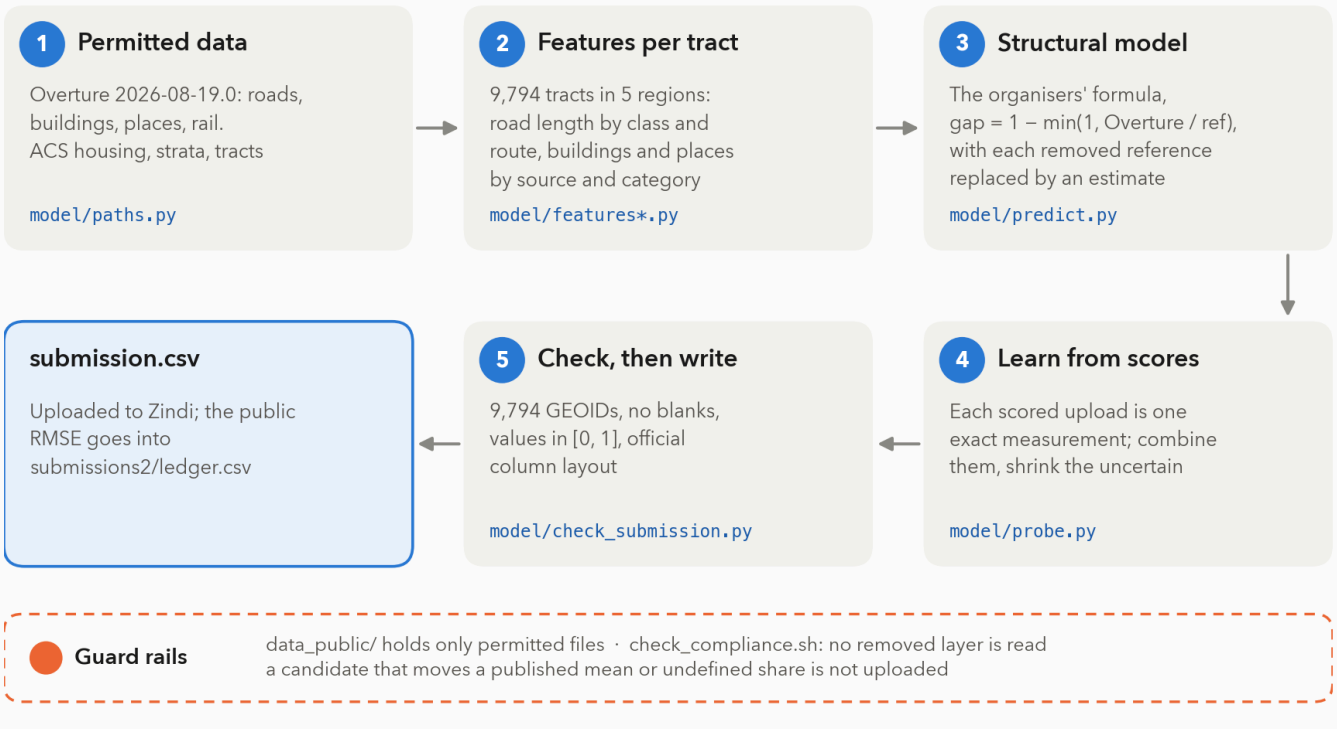


Fig. 1. Steps 1–3 are deterministic; step 4 is the only one that reads public scores; the guard rails run before every upload.

In one minute

The organisers score every census tract with a *coverage gap*: how much of the real world is missing from Overture Maps. On 25 September they removed the reference layers that the score is computed from (TIGER roads, Microsoft buildings, HIFLD/USGS facilities, County Business Patterns). So the task became: **predict a formula's output without the data the formula needs.**

Our solution has two layers, and this document explains both with pictures:

1. **A structural model.** We keep the organisers' formula exactly and replace each removed quantity with the best estimate we can build from the permitted layers. We pin every free parameter to a number the organisers published.
2. **Learning from our own scores.** There are no labels, but every upload returns one exact number. One scored file tells us exactly how far the truth lies in the direction of that file. We combine many such measurements and shrink the uncertain ones.

1. The pipeline at a glance

Fig. 1 (above) shows the whole pipeline. Each step reads only the output of the step before it. Steps 1 to 3 are deterministic and need no leaderboard. Step 4 is the only step that uses public scores. Step 5 refuses a file that Zindi would reject.

2. What the score is made of

The score of a tract is the **mean of its defined components**. Each component compares Overture with a reference: `gap = 1 − min(1, Overture / reference)`. A component is *undefined* when the reference is zero (for example, a tract with no highway in the reference). The organisers published the mean of each component over the scored tracts:

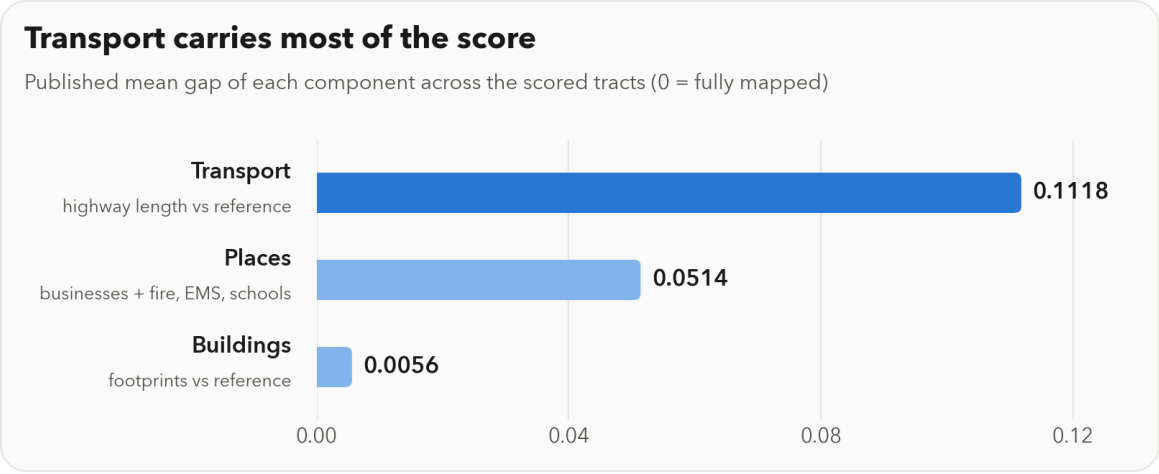


Fig. 2. Most of our effort went into estimating the removed highway reference; buildings are almost always fully mapped.

3. Step by step

Step 1 — Permitted data (`model/paths.py`)

Only files that are still in the organisers' bucket: Overture release 2026-08-19.0 (`buildings` , `roads` , `roads-unfiltered` , `pois` , `rail` , `infrastructure`), ACS housing units, the strata tract tables, tract polygons and the per-region sample submissions. `model/paths.py` lists them and refuses any other path. The pipeline runs from `data_public/` , a folder that contains only those files, so it **cannot** read a removed layer by accident.

Step 2 — Features per tract (`model/features*.py`)

For each of the 9,794 tracts:

- **Roads.** Length inside the tract, measured in metres after projecting to EPSG:5070 and then clipping. The length is split by road class and by route network (Interstate, US, state, Texas FM/RM, county, BIA).
- **Buildings.** Footprint counts by centroid, split by source (Microsoft ML Buildings, OpenStreetMap, Esri).
- **Places.** Counts in total, per facility category (fire, EMS, the six school codes) and per source dataset.
- **Context.** Population, urban share, land area, RUCA, tribal overlap and SVI, from the strata table.

Step 3 — Structural model (`model/predict.py`)

We keep the organisers' definitions and replace each removed reference quantity:

component	removed reference	our estimate from permitted data
transport	TIGER primary and secondary roads	Overture length on numbered routes + 0.17 × un-routed highway-class length (+ 0.56 × county routes in eastern Oklahoma)
buildings	Microsoft footprints	Overture's Microsoft-sourced footprints + the OSM/Esri ones that hide a Microsoft footprint
places	County Business Patterns, HIFLD/USGS facilities	BrightQuery-sourced places; the probability that a reference fire station, school or EMS exists, given what Overture shows

How the free numbers were set. The organisers published the share of tracts with an undefined component in each region. Our transport rule has to reproduce those shares, and it does:

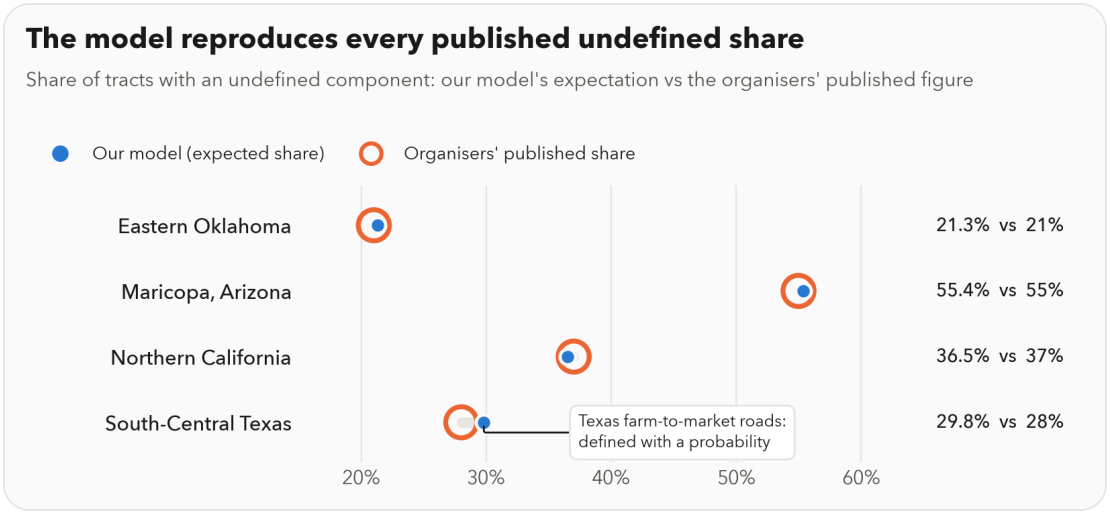


Fig. 3. The model value is the expected transport-undefined share (METHODOLOGY.md, table B.4).

The second published fact is the band of the Overture-to-reference highway ratio (0.71 to 1.59). The weights 0.17 and 0.56 are the values that put every region inside that band. The full reasoning, including boundary highways and the expected-gap formula, is in `METHODOLOGY.md`, Part B.

Step 4 – Learning from our own scores (`model/probe.py`)

There are no labels. But each upload returns its public RMSE, which is the distance between our file and the hidden truth. Three distances are known: the base file's score, the new file's score, and the size of the change we made. Like a triangle with three known sides, this gives **one exact number**: how far the truth lies in the direction of our change.

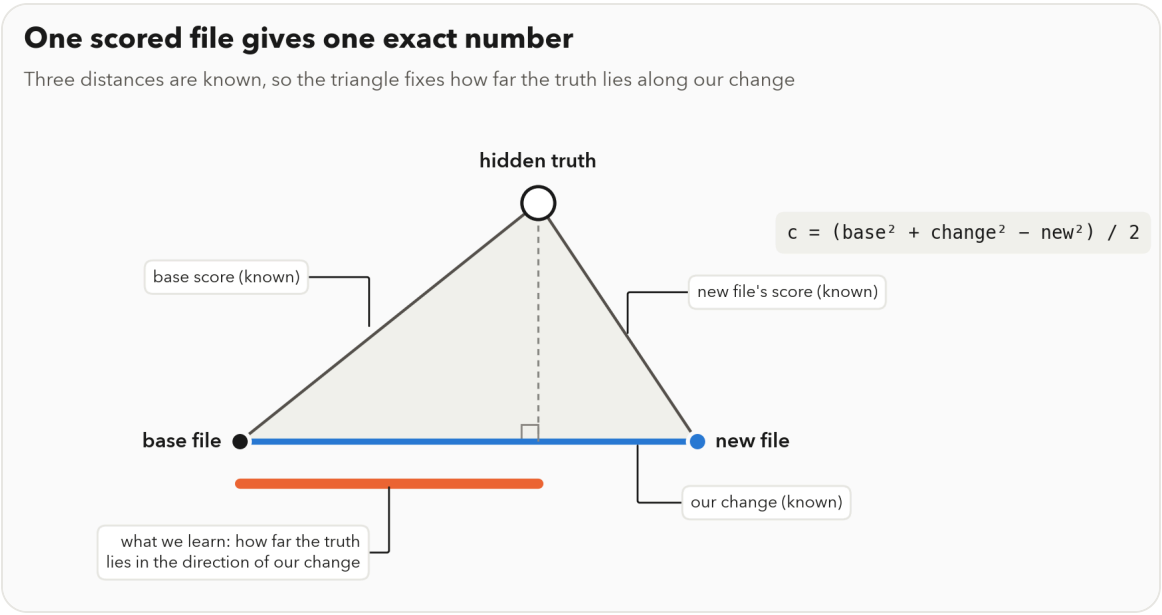


Fig. 4. With mean squared distances, c is the public-tract average of change × (truth − base).

Many scored files give many such directions. `model/probe.py` combines them into one step towards the truth (least squares over the measured directions). Two safety rules keep it honest:

- **Shrinkage.** A measured direction is kept only where it stands clear of the noise that a random public/private split would produce (a positive-part James-Stein factor per direction, set by `kappa`).

- **Published means.** Region and group means posted on the Zindi forum pull each region part of the way towards its measured value (`calibrate_regions` , `calibrate`). Small regions move less.

The public score improved step by step as each idea was measured:

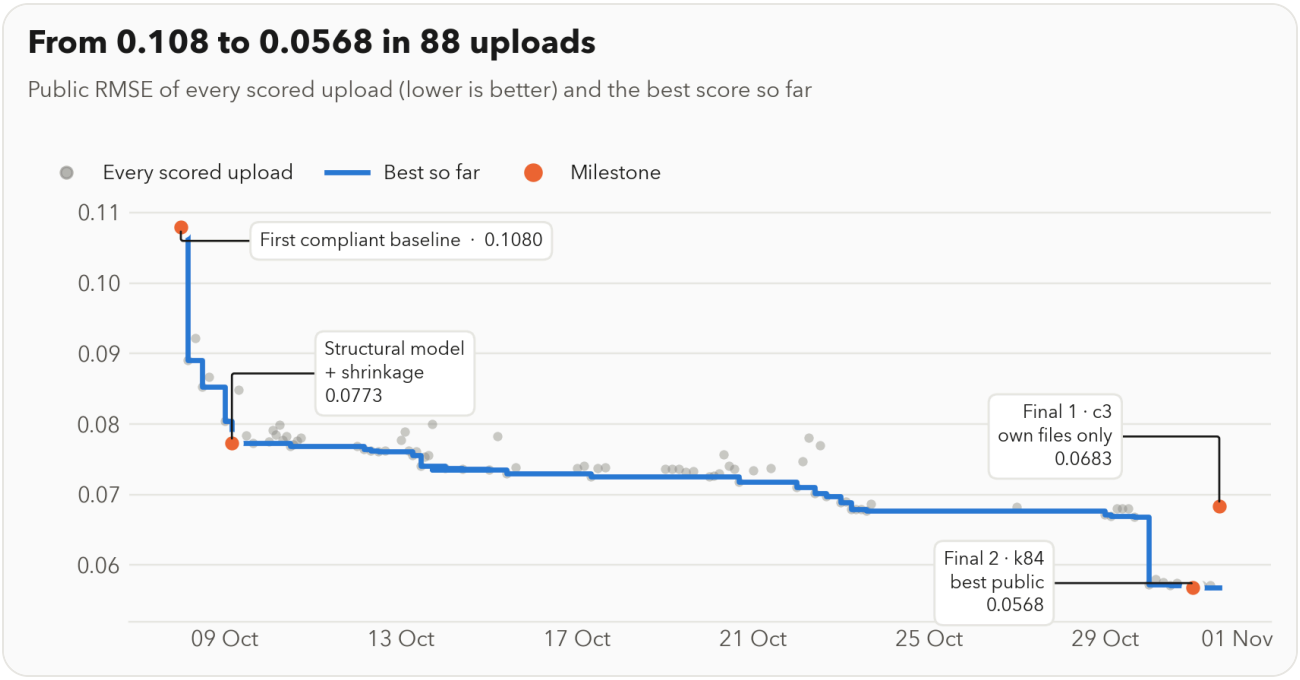


Fig. 5. Grey: every scored upload; blue: the best so far; orange: milestones and the two final files.

Step 5 – Check, then write (`model/check_submission.py`)

Every file is checked before upload: the header is `GEOID,coverage_gap_score` ; the GEOIDs, read as text, are exactly the 9,794 IDs from the sample submissions (no duplicates, no extras); there are no blank cells; and every score is a finite number in [0, 1].

4. The two final files

	`c3` (RzkTLsFj)	`k84` (2gyMPZPa)
public RMSE	0.068339	0.056774
built from	the challenge package, our own scored files, numbers posted on the Zindi forum	starts from a scored file another team published (iam-dante `tr_u1_000`) and also fits on other teams' published files and scores
how	`research/clean_fit.py` + `probe.calibrate_regions` + `probe.calibrate(0.5)` ; one command, byte-identical rebuild (section 5)	`research/bag_fit.py --own --nb 512 --kaps 3 --build 3` gives the fitted step, then a line step along `k53` (`research/LOOP_LOG.md` , 31 Oct); research code, not packaged as one command

We chose one file of each kind on purpose. `c3` uses nothing outside our own work and the public forum, and it is the file this documentation is built around. `k84` scores better but depends on other teams' public files. We

disclose that here so the organisers can apply the rules as they see fit.

5. How to reproduce

```
make env                                          # Python 3.10, pinned packages (requirements.txt)
BB_DATA=data_public python -m model.features      # step 2: per-tract features (~5 min on a laptop)
python -m model.predict                           # step 3: structural model + one-knob variants
./check_compliance.sh                             # guard rail: model/ reads no removed layer
PYTHONPATH=. python3 research/clean_fit.py --out c1_clean_fit                 # step 4
PYTHONPATH=. python3 -c "from model import probe as pr; \
  pr.calibrate_regions('c1_clean_fit', 'c2_clean_regions'); \
  pr.calibrate('c2_clean_regions', 'c3_clean_regions_grp05', 0.5)"
PYTHONPATH=. python3 -m model.check_submission submissions2/c3_clean_regions_grp05.csv   # step 5
```

Step 4 reads the scored files in `submissions2/` and their public scores in `submissions2/ledger.csv`. On 1 November 2026 these commands rebuilt `c1`, `c2` and `c3` **byte-identical** to the uploaded files (sha1 of `c3`: `c781cf25…`) in about 5 seconds.

6. Runtime, environment and performance

item	value
machine	Apple Silicon laptop, 16 GB RAM, no GPU
software	Python 3.10.12, numpy 2.2.6, pandas 2.3.3, pyarrow, DuckDB 1.5.5 (`spatial` , `httpfs`), matplotlib for figures
features (step 2)	about 5 minutes from a local copy; Texas (4.4 M footprints) dominates
fit and calibration (step 4)	about 5 seconds
randomness	none in steps 1–3 and 5; step 4 bagging uses a fixed seed (`--seed` , default 11)
output	rows sorted by GEOID, so a rebuild is byte-identical

7. Error handling and logging

- **Published numbers as a gate.** `model/constraints.py` scores every candidate structure against all published aggregates at once (undefined shares, component means, the 0.71–1.59 ratio band). Candidates that moved a published mean, or an undefined share by more than 3 points, were not uploaded.
- **Input whitelist.** `model/paths.py` raises an error on any path that is not a permitted file.
- **Output check.** `model/check_submission.py` exits with an error and prints the reason (wrong header, missing or extra GEOIDs, blank cell, value outside [0, 1]).
- **DuckDB pitfalls are fixed at the source.** `model/paths.py` sets path-style S3 URLs (the bucket name contains dots) and `model/features.py` projects with `always_xy := true` (otherwise lengths come back as `inf` or `NaN` without an error).

- **Every upload is logged.** `submissions2/ledger.csv` holds the file name, date, public RMSE and a note for all 88 scored uploads; `research/LOOP_LOG.md` records each decision and its evidence.

8. Maintenance and monitoring

- **Pinned inputs.** Overture release 2026-08-19.0 in the organisers' bucket. If the organisers re-cut the data, the published-number checks fail loudly.
- **Schema watch.** The places features need `categories.primary`, which Overture deprecates in later releases. A newer release needs that column mapped before step 2 is re-run.
- **New regions.** Add the region to `REGIONS` in `model/paths.py`; the feature code is region-agnostic.
- **Re-fitting.** After new uploads, append their scores to `ledger.csv` and re-run step 4. Keep the uploaded bytes unchanged: step 4 assumes each file on disk is exactly the file that was scored.

9. Weightings

We use the organisers' equal weighting as specified: the arithmetic mean of the defined components, and inside places the equal mean of the two halves and of the defined facility types. No alternative weighting was applied.

10. What we learned (limits)

- **Published scores are not always exact.** Several teams published files with their scores. When we checked them against our own exactly-known files, some disagreed by 1–5% of a step. Small errors like this can make a combined fit promise far more than it delivers: our `k84` delivered about 13% of its predicted gain from those files. We now model this error (`--cerr` in `research/bag_fit.py`); with it, no further safe gain was found.
- **The public split is 30% of tracts.** A gain on the public board can partly be noise. The shrinkage factor `kappa` and the private-score simulation in `research/bag_fit.py` exist to guard against this.
- **The structural model alone is weak at tract level.** With simple shrinkage it reached 0.0773; most of the later gain comes from step 4, which needs scored uploads.

Full technical detail: `METHODOLOGY.md` (Part B) and the code. Every number above comes from a file in this repository.